GW-1
SOLAR // WRAITH

POCKET GAMMAWAVE · SOUND · FX · MOD

engines, effects, modulation, sequencer and songs

firmware v1.1-GW · M-VAVE FM-1 · gammawaveaudio.com · shadowtune.itch.io/gammawave · fvalente.github.io/GammaWave-FM1

ENGINES · EDIT 1 / EDIT 2

EIGHT ENGINES, ALL SYNTHESIS

ENGINE	EDIT 1				EDIT 2				WHAT TO KNOW
ANALOG	WAVE	DETUNE	MIX	NOISE	CUT	RES	DRIVE	INT	SAW SQR TRI SIN PWM **SYNC RING SAW3**; INT: osc 2 ±24 st. Its filter is per voice.
FM6	ALG	FB	MOD	RATIO	MEG	VMOD	DETUNE	SLOT	Dexed's 6 operators, 32 algorithms drawn, .syx patches (SLOT), MOD / RATIO move the modulators. **OPERATOR · OP ENV · OPERATOR 2**: each operator on the device (ratio / fixed, fine, level, its envelope, mode, detune, velocity). **SLOT B1–B32**: a 32-voice .syx sent from the web editor.
PHASE	WAVE	WAVE2	DCW	ENV	DETUNE	LINE	SUB	FB	Phase distortion (CZ); **FB** feeds the output back into the phase: hard edge to growl.
LOFI	CHIP	WAVE	DUTY	CRUSH	–	VIB	BEND	MASK	Chip: PLS TRI SAW NOIS WRAM. **BYTE**: ALGO VAR · MOTN GRIT BEND LOOP. **FLOAT** (floatbeat): ALGO VAR · MOTN FBK BEND LOOP; CHIP sets FULL / 4BIT / 1BIT.
VOICE	VOWEL	VOWEL2	TALK	SHIFT	BUZZ	BREATH	Q	RAND	Formant voice (after klattsch): vowels, talk, breath.
WHEEL	REG	SUB	BODY	TOP	PERC	CLICK	DRIVE	ROTOR	Drawbar organ: registrations, percussion, rotary.
NOISE	MODE	COLOR	FREQ	PEAK	TRK	DENSE	DRIFT	CRUSH	Coloured and metallic noise, tracked or free.
DRUM-X	MORPH	TUNE	TONE	DECAY	NOISE DRIVE	FM	ENV	FOLD	8 sounds BD SD CP CH OH TM RS CB, sides A / B: **MORPH** morphs the kit. The kit first: **KIT 3** BP FEEDBK, **X-MOD** RATE DEPTH SHAPE RAND. Then a sound (KNOB 1 the sound): SIDE PITCH PMOD · DECAY NOISE COLOR · MODE PM.TYPE DRIVE · **LAYER** (no note) FEEDBK **SEND** (CHO / DLY / REV). Last LEVELS and **STEREO** IMAGE (a knob through the kit's stereo placings).

SPECTROGRAM FILTER & WRAITH BAR

TYPE	CUT	RES	DRV

LP · HP · BP · COMB · VOWEL · SCREAM, drawn over the audio-reactive spectrum. The Wraith ribbon follows CUT through six zones (**FLU · WHI · VOX · CHR · BEL · SCR**; vowels **OO · OH · AH · EH · EE** with VOWEL / SCREAM or the VOICE engine). **OCT– + ARP** steps through 6 Hearth macros.

VOICE · PLAYING

VCE	GLD	GLMOD	PRIO

POLY · MONO · LEG · UNI; glide and its mode; LAST / LOW / HIGH priority. **8 voices** shared by the six tracks (MONO and LEGATO keep their note).

49-GENRE GENERATOR & DICE

OCT– + PLAY (or SELECT on HOME): 49-Genre Song Generator including **TRAP ATL**, **UK DRILL** (3-3-2 tresillo bounce), **DARK TRAP** (808 cowbells), **RAGE TRAP**, house, techno, IDM, ambient… KNOB 1 genre, SELECT its pages (SONG: tempo, key, scale, length; FEEL: space for a voice, fills, wet, grit), OCT+ makes NOW, SAVE keeps it as a **take** (hold OCT+: the keys remake a part, a sound, a section; hold OCT–: the keys play the takes; JAM; SET; share codes). **OCT– + ENV**: 1-click **SEQ > MUTATE** (ORBIT / DRIFT; on the page OCT– FREEZE, OCT+ KEEP). **OCT– + GLO**: Undo.

MODULATION

ENV · LFO

ENV

ATK	DEC	SUS	REL

ENV DEST

FLT	PIT	SHP	LOOP

LOOP: held, the envelope cycles from the sustain: an ADSR-shaped LFO.

LFO

RATE	WAVE	PHS	FADE

SIN TRI SAW SQR S&H

LFO DEST

PIT	FLT	SHP	AMP

MOD MATRIX · 4 SLOTS A TRACK

SLOT	SRC	DST	AMT

SOURCES

LFO ENV VEL KEY RAND MODW AT EXPR M1 M2 M3 M4 STEP

DESTINATIONS

PITCH CUT SHP AMP PAN DIST CHO DLY REV RATE VIB E1…E8 CLIP PNCH

E1…E8: the engine's own EDIT knobs. CLIP / PNCH: the master.

MSEQ · MACROS

MSEQ · A CV SEQUENCER PER TRACK

STEP	LVL	LEN	SLEW

16 levels, DIV on MSEQ 2, in time with the transport. Use it from MOD, SRC STEP.

MACROS M1–M4

Hold LFO: the knobs are M1–M4, latched until LFO again. Each session starts with 2 random routes a track; F3 rolls new ones, G3 clears. By hand: MOD, SRC M1…M4. Saved with the project.

SEQUENCER

64 steps a track, up to 4 notes a step, piano roll and drum grid. **Hold a step + turn a knob**: a parameter lock. Step recording: hold a chord + KNOB 1 its length; + ARP: arpeggiated (K1 rate, K2 order, K3 oct). COMP / COMP 2: two components a step (CHANCE, RATCH, ARP NOTES / SEQ / SCALE, TRIG, RAND, OCT, TONAL, BEND, ECHO, PULSE, JUMP, REV: TIMES x1 x2 1:2, FILL, KILL); a TRIG or FILL wins; RATCH / ECHO: DIV; WHEN: each one's loops (ALWAYS 1:2 2:2 SKIP) and the step's LEN. SHIFT's CHANCE thins the whole track; C.LEN gives the components a length of their own (they drift over the steps). REC + knobs while playing: automation (64 records a section, MOTION page plays / clears them, AUTO LIST edits them).

PATTERN · SHIFT

LEN	DIV	SWG	GATE
OFS	PIT	CHN	CLEN

SCALE · ARP

ROOT and SCALE are every melodic track's: CHR MAJ MIN DOR MIX PEN MPEN HARM PHRY LYD LOC MEL BLUES WHOLE DIMHW DIMWH. QNT snaps keys or the sequence, TRN transposes, **ALL TR** (SCL 2) moves every sequence along the scale. **CHORD**: a key plays a chord (TRIAD 7TH ADD9 SUS2 SUS4 in key, MAJ MIN 7 MAJ7 MIN7 POWER), VOICING LEAD leads each to the next, STRUM as a harp.

ARP MODES

OFF UP DN UPDN RND ORD REPEAT DNUP UP+8 CONV DIVG PINKY THUMB WALK CHORD **TRNS** (the keys transpose the sequence)

SONGS

8 projects × 4 sections (A–D), a song of up to 16 rows by bars (SAVE › ARRANGE; END: STOP, x2–x8, LOOP), SLOOP style. Each row can hold a **scene**: mutes, DRUM-X group mutes, macros and MIDI punch-ins. SAVE held plays sections live on the next bar, stores them, records the song (SONG REC). Each section keeps its sounds, steps, motion, kit and its SLICER.

EFFECTS

PER TRACK

FX · SENDS

DST	CHO	DLY	REV

DIST

DST	TYPE	TONE	–

SOFT · HARD · FOLD · CRUSH · RECT.

AUTO PAN

DEPTH	RATE	SYNC	WAVE

The track's pan swung L–R around its PAN. SYNC **BEAT**: RATE 4 BARS … 1/32, in time with the bar; **FREE**: in Hz. SIN · TRI · SAW · SQR · S&H, never a click.

SLICER · ONE ON A BUS

SLCR	PAT	RATE	DEPTH

OFF · GATE · STUT, 16 patterns, 1/8 … 32T. **SL BUS**: T1–T6 and **FX** (the delay / reverb / chorus returns) ON / OFF: everything on = the whole mix chopped after the effects. **SL PITCH**: the repeats' 16-step pitch (±24 st, tape style, QNT SCL to the scale).

DELAY · CHORUS

DLY

TIME	FDBK	TONE	MIX

DLY 2

WIDE	PITCH	SPRY	MOD

DLY 3

TYPE	PING	–	–

Synced or free (5 ms – 1.48 s); PITCH / SPRY: a grain delay.

CHORUS

RATE	DEPTH	STER	–

STER: 0 as always, down to mono, up to twice as wide.

REVERB · FIVE MODELS

TYPE	SIZE	DAMP	PRE
MOD	RATE	FILT	WIDE

TYPE	WHAT IT IS	SIZE	DAMP	MOD	RATE	WIDE
ROOM	Classic room	decay	darkens	chorus on the tail	its speed	stereo
SPRING	Amp spring tank	length · decay	darkens	wobble	wobble speed	stereo
SHIMR	Up an octave on every pass; **PRE → AIR** (±24 st: down darker and lower, up airier and higher)	decay	darkens	how much shimmer	—	—
RESO	4 strings tuned to the scale; **PRE → PITCH** (0–14 steps up), **DAMP → FOLW** (OFF / T1–T6: the chord follows that track's notes)	how long they ring	FOLW	past half: the chord opens	—	—
CLOUD	Granular wash; SIZE max **freezes** it; **PRE → AIR** (±24 st: the cloud down / up, darker / airier)	decay · freeze	darkens	grains pitched + chorus	grain density	grain spread

MASTER

CLIP	PNCH	DUCK	REL

CLIP saturates without getting louder. **PNCH** drives the drums: denser body. **DUCK**: the kick pulls the rest down (REL its release). An always-on **leveler** (slow 2:1, −9…+6 dB) before the limiter keeps patches at a steady volume.

PUNCH-IN FX (FX HELD)

REPEAT 1/8 · 1/16 · 1/32, LPF, HPF; on the notes: OCT–/+, ½ TEMPO, DEC–/+, STUTTER, ATK+, ARP, RANDOM. Knobs: FILTER CRUSH THROW DEPTH. Recordable per section.

GPL-3.0 · Companion to GammaWave Desktop VST (gammawaveaudio.com) · Built on Felucca, JIANT & SLOOP